Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Systems Fundamentals I & II, Statistics & Probability

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, and custom date/time picker integration to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

PROJECTS

TuneAcademy React, FastAPI, Docker, GCP, Firestore

- Set up a CI/CD pipeline using GitHub Actions to lint, test, and auto-build/deploy a React/Tailwind frontend and Python/FastAPI backend running in Docker, with a Cloud Firestore database and GCP hosting.
- Built the AI/audio-analysis pipeline for an app matching music students with instructors, using Essentia and NumPy to extract pitch, rhythm, and tone features and rule-based scoring algorithms to detect technical weaknesses.

Diary Handwriting Analysis Python, PyTorch, Hugging Face Transformers, OpenCV

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV preprocessing pipeline, reaching ~87% reliable transcription, then pivoted to DANIEL, a segmentation-free architecture for joint transcription and named-entity extraction to surface mood and topic patterns across diary entries.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University Sep 2025 – Dec 2025

- Built a pipeline using Python and Selenium to monitor newly issued SSL/TLS certificates and flag fraudulent domains impersonating EZ-Pass tolling services, designing heuristic-based detection logic to triage hundreds of flagged domains.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original midterm review materials adopted as the basis for later semesters and hosted review sessions reaching 100+ students.

International Peer Mentor, Office of Global Affairs, Stony Brook University Aug 2025 – Present

- Mentored 6 international students through the university's International Student Success program via biweekly one-on-one meetings, providing guidance on cultural adjustment, academic strategy, and interpersonal conflict resolution.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL (SQL), AWS, GCP, Firebase/Firestore, MongoDB, Supabase

Tools: Git, Docker, Linux, PM2, Selenium